

The Accreditation Problem: Continuous-Validity Authorization for Agentic Systems (v1.2 cover-branding restoration)

Justin H. Kuiper, CISSP

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Abstract

The authority to operate a system that processes classified information — the accreditation that an organization formally recognizes as the authorization for that operation — is, in the enterprise information security model, a point-in-time determination: an authorizing official examines the system, evaluates its security controls against a defined baseline, and issues an authorization to operate (ATO) that remains valid for a specified period. The ATO is a governance artifact produced at a known moment and presumed valid until its expiration or revocation. This model fails at the tactical edge under DDIL conditions not because the ATO was incorrectly issued, but because the model assumes a stability of operational conditions that DDIL environments do not provide: the system that was accredited may be physically different from the system currently operating; the security controls that satisfied the accreditation baseline may have drifted; the authority that issued the ATO may be unreachable for verification; and the node may have received software or configuration updates during a connectivity window that altered the security posture the ATO was based on.

This paper argues that accreditation at the tactical edge is not a point-in-time determination — it is a continuous-validity problem. The question that matters operationally is not "was this node ever accredited?" but "is its accreditation currently valid under the operational conditions now obtaining?" Answering that question under DDIL conditions requires treating accreditation state as a distributed system property — tracked in the WAL, replicated via the CRDT layer, carried through DTN custody chains, and continuously assessed by the AI Supervisor against an accreditation model trained on the operational history of the substrate.

The paper develops four structural failure modes in the enterprise accreditation model when applied to DDIL operations — accreditation staleness, physical accreditation drift, revocation propagation delay, and AI Supervisor accreditation boundary failure — and proposes a substrate architecture in which these failures are addressable without abandoning the foundational principle of human-governed authorization. The §6 Governor Application grounds this architecture in HGC³AE²: accreditation boundary decisions belong to the H half; accreditation state cataloging and Supervisor curriculum belong to the C³ half; accreditation verification at custody and boundary crossings belongs to the AE² half.

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Abstract

The accreditation model that classified-information systems have inherited from NIST SP 800-37 and adjacent doctrine is a **snapshot** model: a system is accredited at a point in time, on the basis of evidence gathered up to that point, and the accreditation remains valid until the next scheduled review.¹ For agentic systems on the tactical substrate (EDO-P1), the snapshot model is structurally inadequate. The system evolves continuously: fine-tunes apply, personas update, envelopes adjust, substrate state drifts. Accreditation must therefore become **continuous-validity authorization**: an architectural commitment that the accreditation evidence is generated, maintained, and verified continuously, and that the accreditation’s validity at any moment is a runtime computable property.²³ This paper specifies the architecture. Existing draft material in `drafts/edo-wave-2/P9-accreditation-problem.md` informs the present draft.

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1. Introduction

NIST SP 800-37 established the Risk Management Framework (RMF) as the dominant accreditation methodology for U.S. federal and DoD systems.⁴ The framework’s value is well-established; its application to agentic systems requires translation, particularly under DDIL conditions where the snapshot model’s underlying assumptions weaken. Continuous-monitoring extensions (NIST SP 800-137) and continuous-ATO research have moved the RMF

¹National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

²Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

³National Institute of Standards and Technology, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, NIST SP 800-137 (Gaithersburg, MD, 2011), <https://doi.org/10.6028/NIST.SP.800-137>.

⁴National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

toward runtime validity verification.⁵⁶ This paper applies that direction to agentic operation specifically.

§2 names five failure modes of snapshot-accreditation for agentic systems. §3 specifies continuous-validity architecture. §4 treats validity as continuous. §5 binds to Skipjack. §6 returns to HGC³AE².

2. Failure Modes of Snapshot Accreditation for Agentic Systems

Drift between accreditation and runtime. The accredited configuration diverges from the running configuration over time; the accreditation is operationally stale before the next scheduled review.

Fine-tune outside the boundary. A fine-tune applied between accreditations produces a model that has not been accredited; the runtime accreditation status is ambiguous.

Persona-update without re-accreditation. Persona prompt, scaffolding, or tool-manifest changes alter the system’s behavior in ways the accreditation evidence did not address.

Envelope-expansion ambiguity. When the governance layer expands the envelope, is the accreditation still valid? The snapshot model has no clean answer.

Reconnection-evidence gap. When a substrate reconnects after partition, the accreditation evidence accumulated during partition may not have been transmitted; the receiving accreditation system reasons against a stale evidence set.

3. Continuous-Validity Authorization Architecture

Accredited-baseline as runtime property. The current accreditation baseline is a runtime-queryable property: every operationally significant component (model, persona, envelope, configuration) carries a reference to the baseline against which it is accredited.⁷

Continuous evidence pipeline. Evidence required for accreditation — configuration state, test results, audit logs, attestation chains — is generated continuously by the operational substrate and aggregated to the accreditation authority. The pipeline does not require periodic batched evidence gathering.⁸

Re-accreditation triggers. Explicit events trigger re-accreditation review: fine-tune deployment, persona update, envelope expansion, substrate reconfiguration. The triggers are enumerated; the agentic system surfaces each event as it occurs.

⁵National Institute of Standards and Technology, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, NIST SP 800-137 (Gaithersburg, MD, 2011), <https://doi.org/10.6028/NIST.SP.800-137>.

⁶Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

⁷Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

⁸Atul Adya et al., “Slicer: Auto-Sharding for Datacenter Applications,” *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI ’16)*, 2016, <https://www.usenix.org/conference/osdi16/technical-sessions/presentation/adya>. (Architecturally relevant evidence-pipeline pattern; supplemented by compliance-as-code research below.)

Authority chain. The accreditation authority (DAA / AO in DoD vocabulary) holds responsibility for each accreditation decision and for re-authorization at trigger events. The architecture surfaces decisions to the authority; it does not auto-accredit.

Compliance-as-code. Accreditation rules are encoded in machine-checkable form (compliance-as-code) so that the runtime can verify configuration against accreditation rules without batched human review.⁹

4. Continuous Validity as Process

Three phases. **Generation** of evidence continuously by the operational substrate. **Verification** continuously against the accreditation baseline. **Re-authorization** on trigger events, with the authority chain explicitly invoked.

5. Skipjack Coupling

Skipjack mechanisms operate over the continuous-validity surface.¹⁰ **Context integrity** is enforced through accreditation baseline verification. **Structured routing** uses accreditation status as a routing input. **HITL control points** fire on re-authorization triggers. **Iterative feedback** closes via accreditation telemetry feeding governance ceremony.

6. Implications and Conclusion

The implications extend to compliance posture (continuous-ATO is operationally feasible and architecturally specifiable), tooling (compliance-as-code and continuous-monitoring pipelines), and authority structures (DAA/AO engagement at trigger events rather than periodic reviews).

The core claim: **accreditation for agentic systems is a continuous-validity property, not a snapshot; the architecture specifies how the property is computable at runtime and re-authorized at meaningful events.**

Returned to HGC³AE² letter by letter:

- **H — Human-governed.** Accreditation authority (DAA/AO) is human; the architecture surfaces decisions to that authority.
- **G — Curated Context.** Accreditation baseline and evidence pipeline are curated context.
- **C¹ — Cybersecurity.** Accreditation is the security-baseline anchor.
- **C² — Continuity.** Continuity of accreditation is the paper's commitment.
- **C³ — Coherence.** Accreditation baseline coherent with current operating configuration.
- **A — Agentially Engineered.** Evidence generation is agentic; authority decisions are human.
- **E — Executed.** Continuous verification at runtime.
- The exponent — the closure — is the generate-verify-re-authorize loop.

⁹Niloofar Reza, Mauro Conti, and Saqib Hakak, "Compliance-as-Code: A Survey," *ACM Computing Surveys* 56, no. 8 (2024): article 200, <https://doi.org/10.1145/3651308>.

¹⁰Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

The remaining EDO papers (interoperability, deployment) operate on the accreditation surface this paper specifies.¹¹¹²

References

— end v1.0-preprint-draft —

Drafting state: v1.0-preprint draft. PR count: 12 inline (NIST SP 800-37 · NIST SP 800-137 · Chaillan IEEE Software · Adya OSDI · Reza ACM CSUR · Kelly DSN · Shahin IEEE Access · Khan C&S · Sandhu IEEE Computer · Havelund ASE · Bloomfield SCS + supporting). Existing draft drafts/edo-wave-2/P9-accreditation-problem.md folds into v0.5 expansion.

Appendix — v1.1 Errata Addendum

Added 2026-05-17 per Round-2 EDITORIAL-STAMP HOLD-substance verdict. The v1.0-preprint remains canonical at its original DOI; this v1.1 supplements with the corrections below.

§X. v1.1 Errata — Round-2 Substance Corrections

§X.1. Why this addendum exists

EDO-P9, *The Accreditation Problem: Continuous-Validity Authorization for Agentic Systems*, shipped as v1.0-preprint on 2026-05-01 at DOI 10.5281/zenodo.20180170, reader-visible on nonsequitur.tech/pubs/white-papers/accreditation-problem/. The paper received its Round-1 editorial stamp (1 of 3) on 2026-05-12.

On 2026-05-17, the Round-2 second-pass review (stamper: Edna; parent thread: YKS-Spine-Binder#355) returned a verdict of HOLD-substance. The review identified four substance defects and a citation/reference cluster issue in the published v1.0 text. Because v1.0 is in the wild with an active DOI, the remediation path is this errata addendum rather than a silent revision. v1.0 remains live and citable; v1.1 ships as a companion record naming what changed and why.

The defects, in summary: (1) §4 is substantively thin — three sentences restating §3 as process phases without operational animation; (2) §5's Skipjack coupling is label-affixed rather than load-bearing — it declares mappings without specifying what continuous-validity changes about each Skipjack mechanism; (3) the [^evidence-pipeline] citation stretches a datacenter auto-sharding paper to anchor an accreditation evidence-pipeline claim without making the architectural analogy explicit; (4) seven references appear in the reference list but are never cited in the body text, inflating the apparent citation count. Each is corrected below.

§X.2. Defect: §4 process-phase thinness

v1.0 said: “Three phases. **Generation** of evidence continuously by the operational substrate. **Verification** continuously against the accreditation baseline. **Re-authorization** on trigger

¹¹EDO-P10 promotion (in concurrent drafting; yks-build#109).

¹²EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).

events, with the authority chain explicitly invoked.” (§4, complete text.)

The problem: §4 restates §3’s five architectural commitments as a three-phase process list without animating how the process executes on the substrate. The section is shape without substance. The Round-2 stamp finding (§4 substance audit) noted the contrast with EDO-P1’s treatment of substrate lifecycle, which walks through concrete operational sequences, and identified two acceptable remediation paths: (a) a concrete operational walkthrough showing what happens at a specific trigger event, or (b) an honest scope note deferring the process animation to implementation-level treatment.

Canonical correction: §4 adopts remediation path (a). The corrected text replaces the three-sentence list with a concrete operational walkthrough anchored to a single trigger event — fine-tune deployment — that traces the generate-verify-re-authorize loop end to end:

§4. Continuous Validity as Process

The architecture in §3 specifies five commitments. This section animates them as a single operational sequence, using fine-tune deployment as the exemplar trigger event.

Generation. The deployment pipeline emits a structured attestation record at fine-tune application: model identifier, training-data provenance hash, resulting weight checksum, timestamp, and the accreditation-baseline version against which the prior model was accredited. This record enters the continuous evidence pipeline (§3) without batched gathering — the pipeline ingests it at deployment time. The attestation schema is compliance-as-code: the fields required are specified in the accreditation-rule encoding, and the pipeline validates completeness before forwarding.¹³

Verification. The accreditation baseline (§3, runtime-queryable property) is read. The new model’s attestation record is evaluated against the baseline’s accreditation rules. Two outcomes are possible: (1) the fine-tune falls within the accredited envelope — e.g., the training data, hyperparameters, and resulting behavior profile match the categories the baseline already covers — in which case the verification step records a pass and the model enters service under the existing accreditation; or (2) the fine-tune exceeds the accredited envelope, in which case verification records a boundary-exceeded finding and the model does NOT enter service. The model is held pending re-authorization.¹⁴

Re-authorization. On a boundary-exceeded finding, the authority chain (§3) is invoked. The accreditation authority (DAA/AO) receives the attestation record, the verification finding, and the specific envelope boundary that was exceeded. The authority renders a re-authorization decision: approve (expanding the baseline to cover the new envelope), reject (the fine-tune does not deploy), or conditionally approve (deploy with additional monitoring constraints). The decision is recorded as a new accreditation-baseline entry, and the baseline version increments. The HITL control point here is not optional — auto-accreditation is architecturally excluded (§3, authority chain).¹⁵

The same generate-verify-re-authorize sequence applies, *mutatis mutandis*, to the other trigger events enumerated in §3: persona update, envelope expansion, and

¹³Niloofar Reza, Mauro Conti, and Saqib Hakak, “Compliance-as-Code: A Survey,” *ACM Computing Surveys* 56, no. 8 (2024): article 200, <https://doi.org/10.1145/3651308>.

¹⁴Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

¹⁵National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

substrate reconfiguration. The fine-tune walkthrough is canonical; the remaining triggers vary in attestation schema but not in process shape.

Why: A paper titled “Continuous-Validity Authorization” must show the authorization process running, not merely name its phases. The walkthrough makes the generate-verify-re-authorize loop testable against a concrete scenario. This is an architectural-specification paper; its sections must specify, not gesture.

§X.3. Defect: §5 Skipjack coupling is label-affixed, not load-bearing

v1.0 said: “**Context integrity** is enforced through accreditation baseline verification. **Structured routing** uses accreditation status as a routing input. **HITL control points** fire on re-authorization triggers. **Iterative feedback** closes via accreditation telemetry feeding governance ceremony.” (§5, complete substantive text.)

The problem: The Round-2 stamp finding (§5 substance audit) identified that each Skipjack mechanism gets one sentence mapping to a continuous-validity concept, but no sentence explains *what changes* about the mechanism when the accreditation surface is continuous rather than snapshot. The section declares a coupling without specifying the delta. Without the delta, §5 is a label, not a specification.

Canonical correction: §5 is expanded to show, for each of the four Skipjack mechanisms, the specific behavioral difference between snapshot-